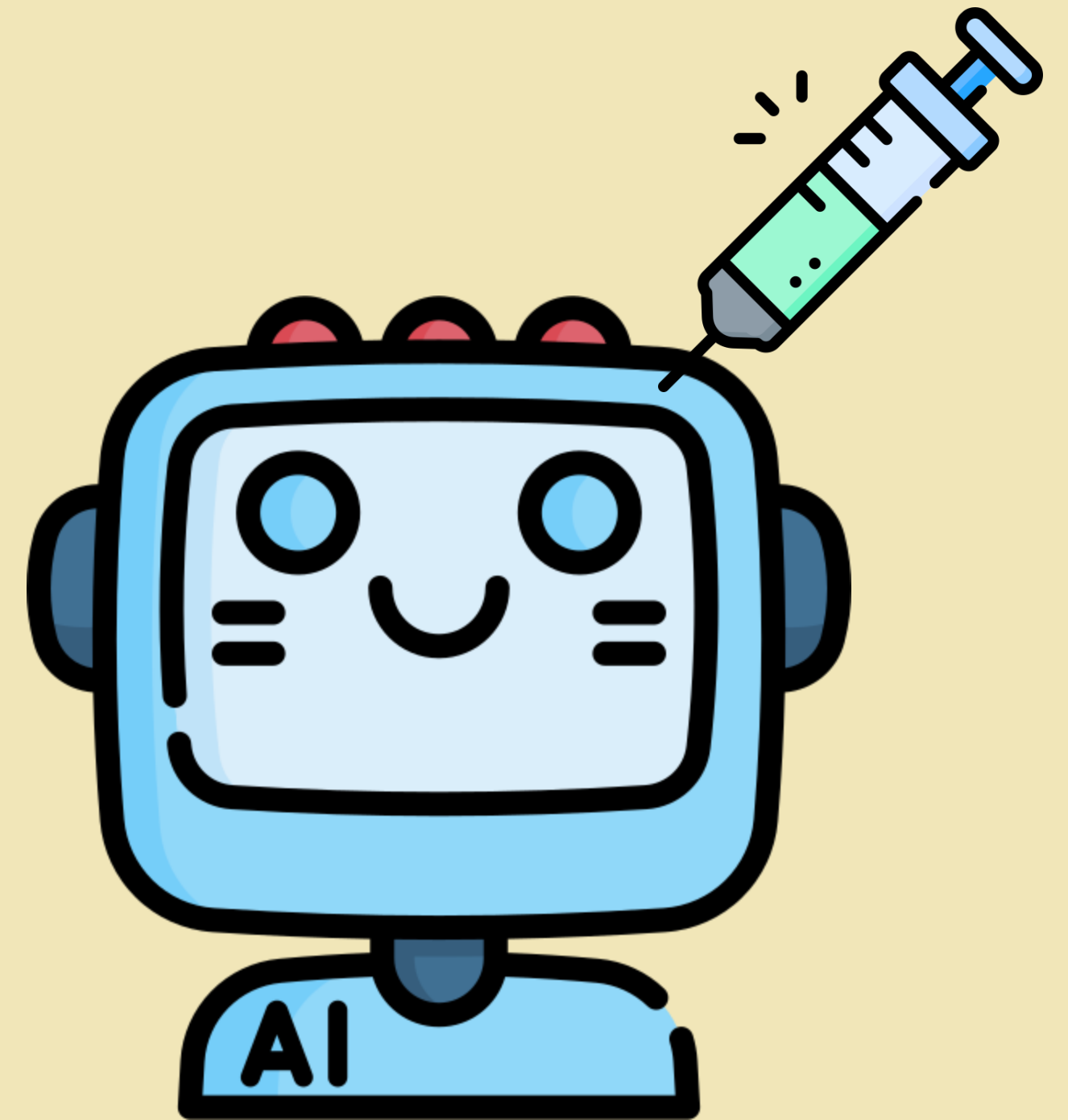




Poisoning Budget Analysis

Nils Pudenz,
52509954

Current Topics in
System Security - AI
& Security
(622.780, 26S)



Poisoning Budget Analysis

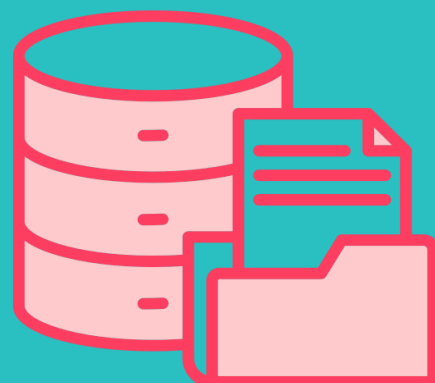


How much data must an attacker
manipulate to successfully degrade
model performance or create targeted
blind-spots?



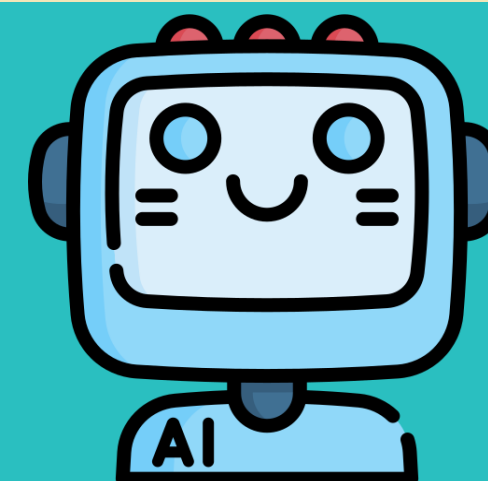


Poisoning



Data

Adversary controls subset
of training data



Model

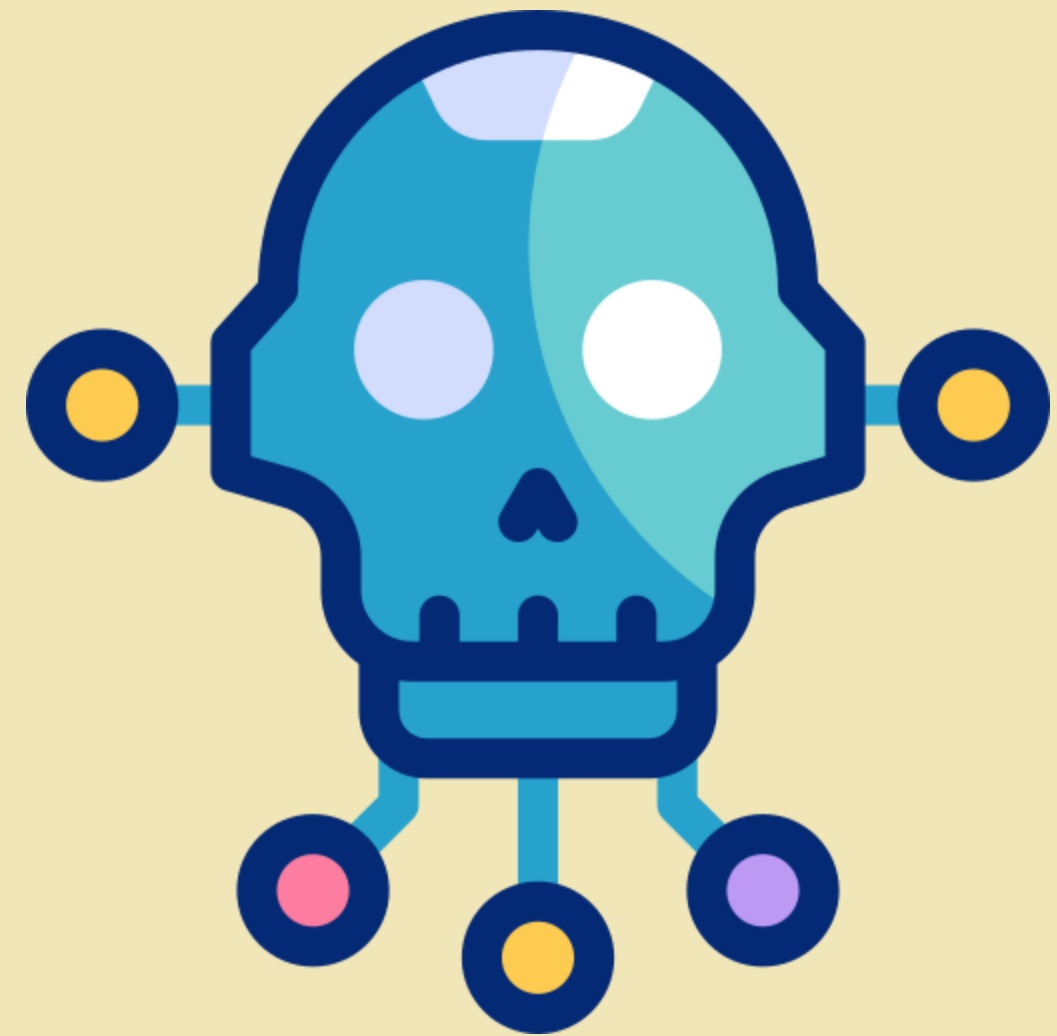
Adversary controls model &
parameters





1 Data Poisoning

Training Time Attacks





Poisoning

Targeted Attacks (Integrity)

induce incorrect predictions on a
specific target set of instances

Reliability Attacks (Availability)

maximize the overall prediction or
decision error



Optimization Problem

$$\min_D R_A(D; S) \text{ s.t. } c(D_0, D) \leq C$$



Optimization Problem

$$\min_D R_A(D; S) \text{ s.t. } c(D_0, D) \leq C$$

Dataset (D_0): 10,000 Emails

Budget (C): 5% $\rightarrow$ 500 Emails

Flip Decision (z_i): 1 if flipped, 0 if untouched

Flip Cost (c_i): 1 (every email costs the same to flip)

General Formula: $c(D_0, D) = \sum_i z_i c_i$

Applied to Enron: $c(D_0, D) = \sum_{i=1}^{10,000} z_i \cdot 1$

The Constraint: $\sum_{i=1}^{10,000} z_i \leq 500$

$$\max_z \sum_{j \in D_{\text{clean}}} \text{loss}(y_j, f_{\text{poisoned}}(x_j))$$

$$\text{s.t. } \sum_{i=1}^{10,000} z_i \leq 500 \text{ and } z_i \in \{0, 1\}$$

The Label (y_i): This is the classification of the email (e.g., 1 = Spam, 0 = Ham).



2

Threat Model

White Box Attack

knowing the feature space F , the true dataset D , and the learning algorithm A





Threat Model

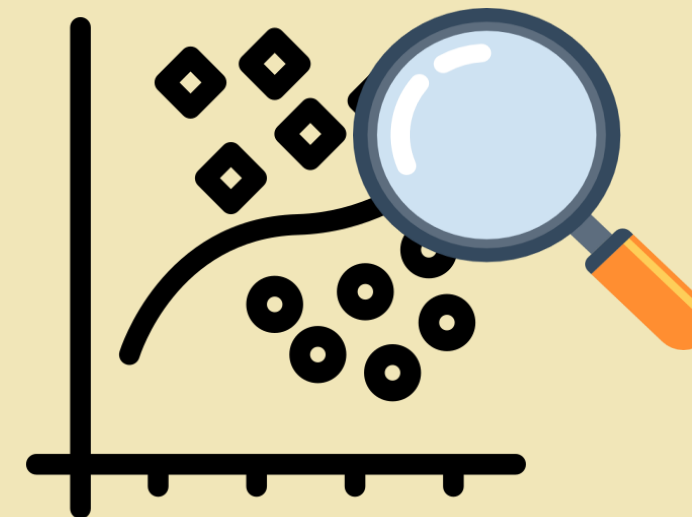


Random &
Targeted Label-
Flipping

Tabular Data
Poisoning



Pipeline



Logistic Regression

Baseline Model (Training
Data without any
changes)

Inserting poisoning

Test data is
manipulated with
poisoned data

Boundary Analysis

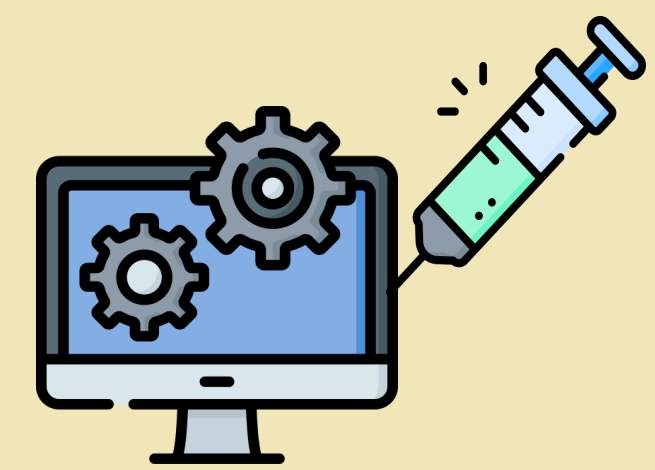
How much poison
do we need to make
the model misbehave

Results

What is the result of
the different trials



Experiments



Binary

- **Spam Email Detection**
(Spam vs. Ham)
- Enron Spam Data

Multi Label

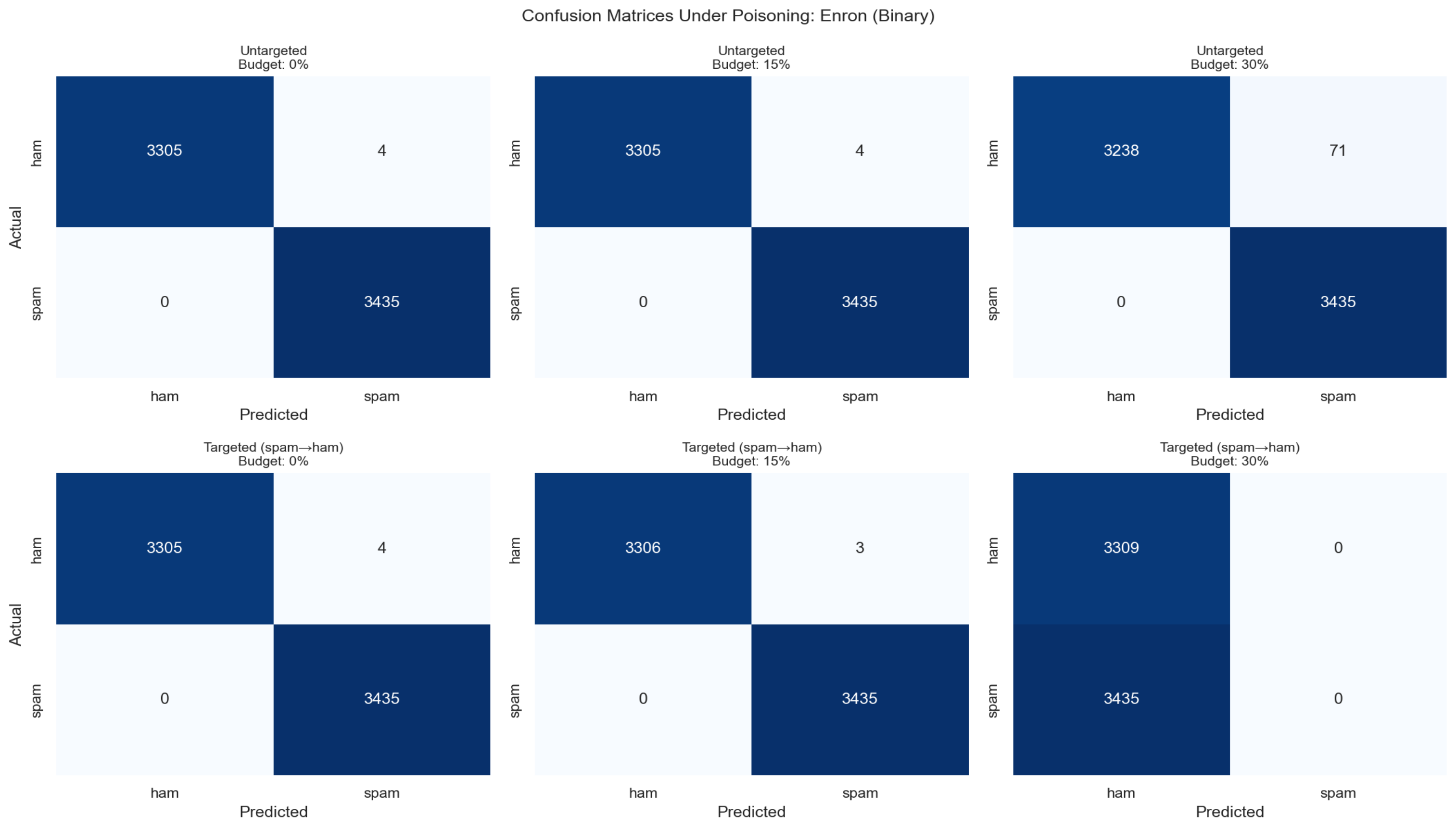
- **Malicious URLs** (phishing, benign, defacement,...)
- Malicious URLs dataset

Tabular Data

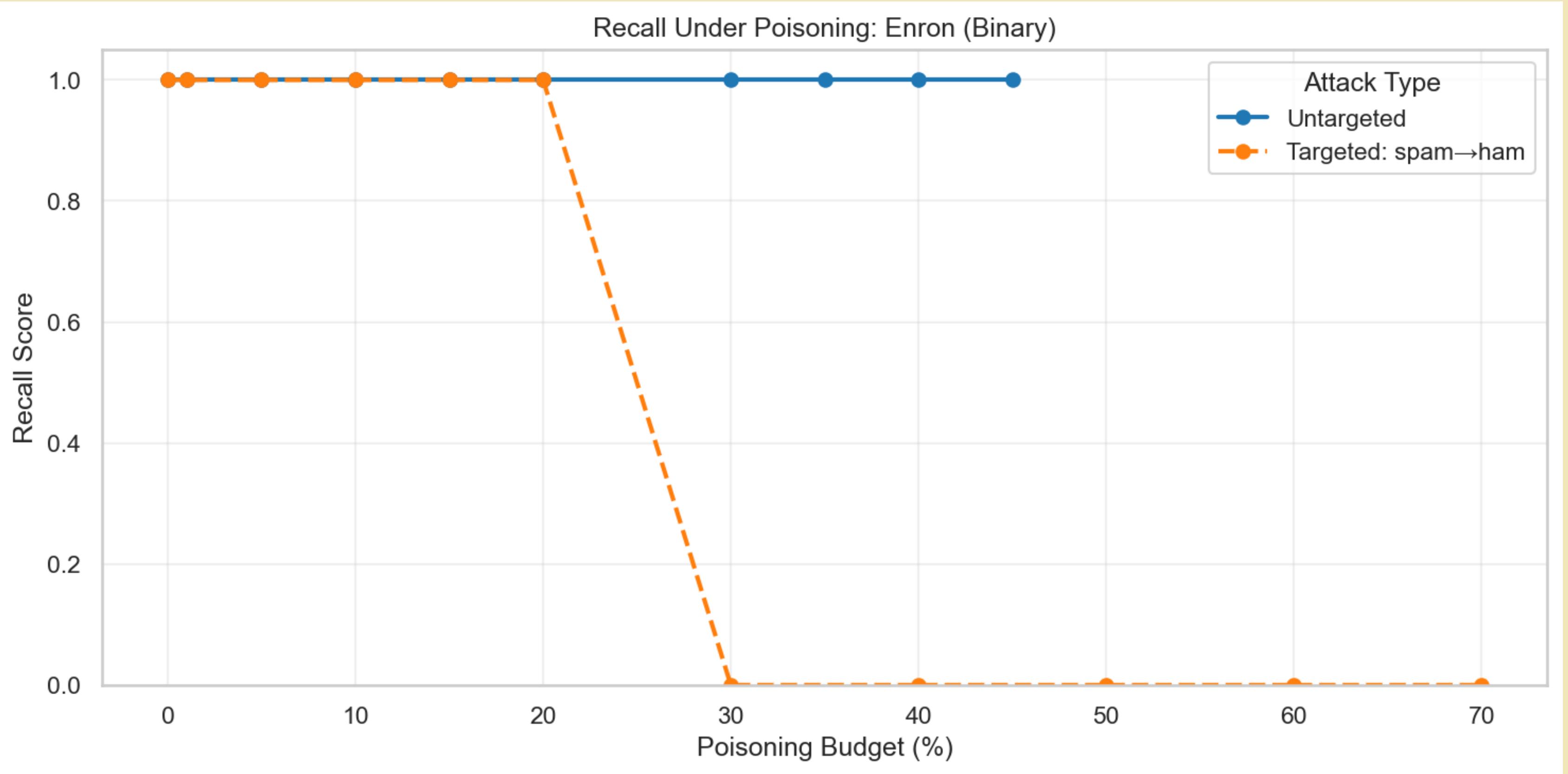
- **Network Activity**
- NSL-KDD



Results - Binary



Results - Binary



Results - Multiclass



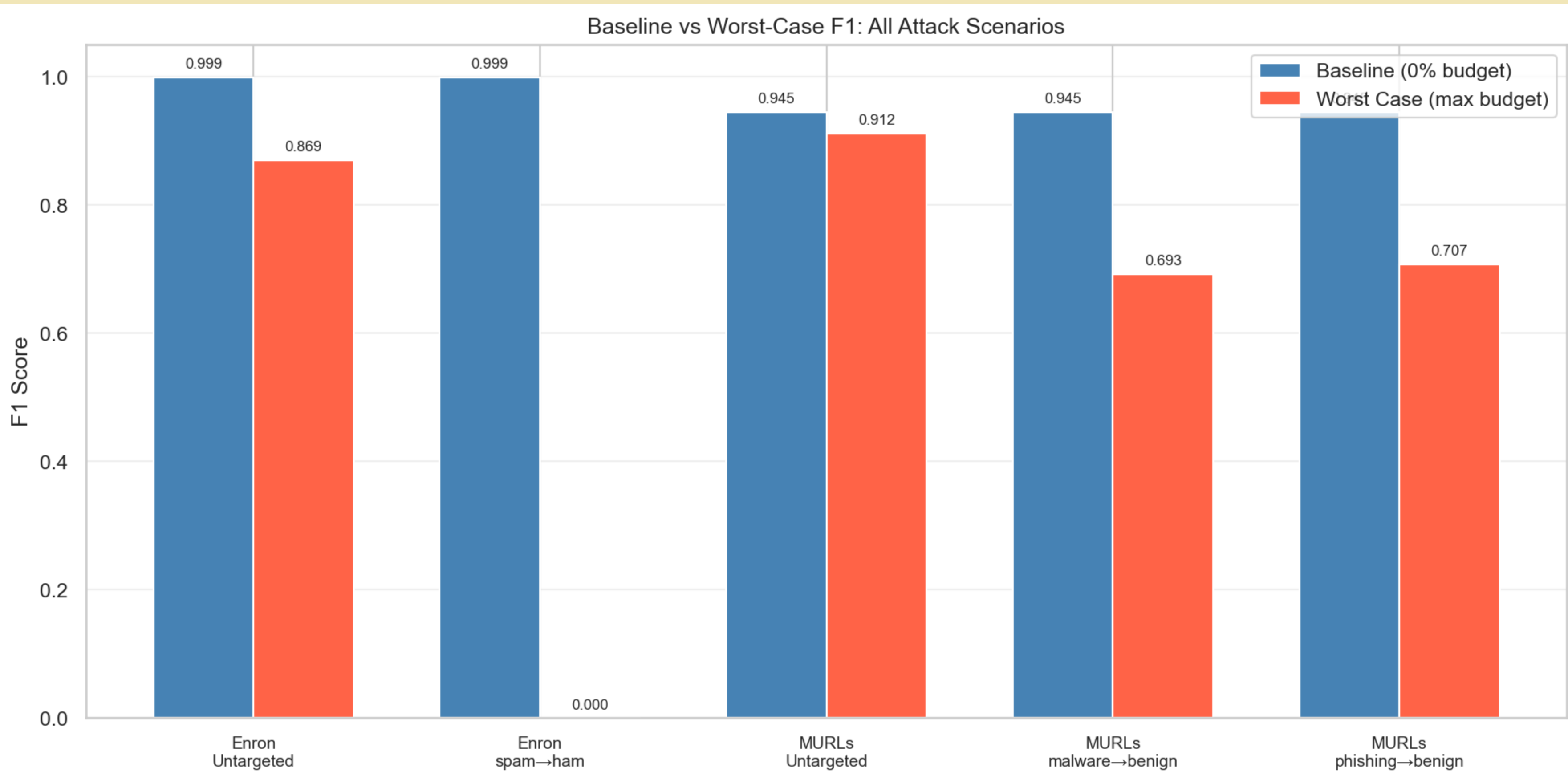
Results - Multiclass

Targeted (phishing→benign) Budget: 0%					Targeted (phishing→benign) Budget: 15%					Targeted (phishing→benign) Budget: 30%							
Actual	benign	defacement	malware	phishing	Actual	benign	defacement	malware	phishing	Actual	benign	defacement	malware	phishing			
	benign	defacement	malware	phishing		benign	defacement	malware	phishing		benign	defacement	malware	phishing			
	Predicted					Predicted					Predicted						
benign	84208	48	39	1326	benign	85532	61	28	0	benign	85532	61	28	0			
defacement	62	19075	7	148	defacement	297	18985	10	0	defacement	297	18985	10	0			
malware	129	64	6037	274	malware	448	63	5993	0	malware	448	63	5993	0			
phishing	2683	285	82	15772	phishing	18404	343	75	0	phishing	18404	343	75	0			

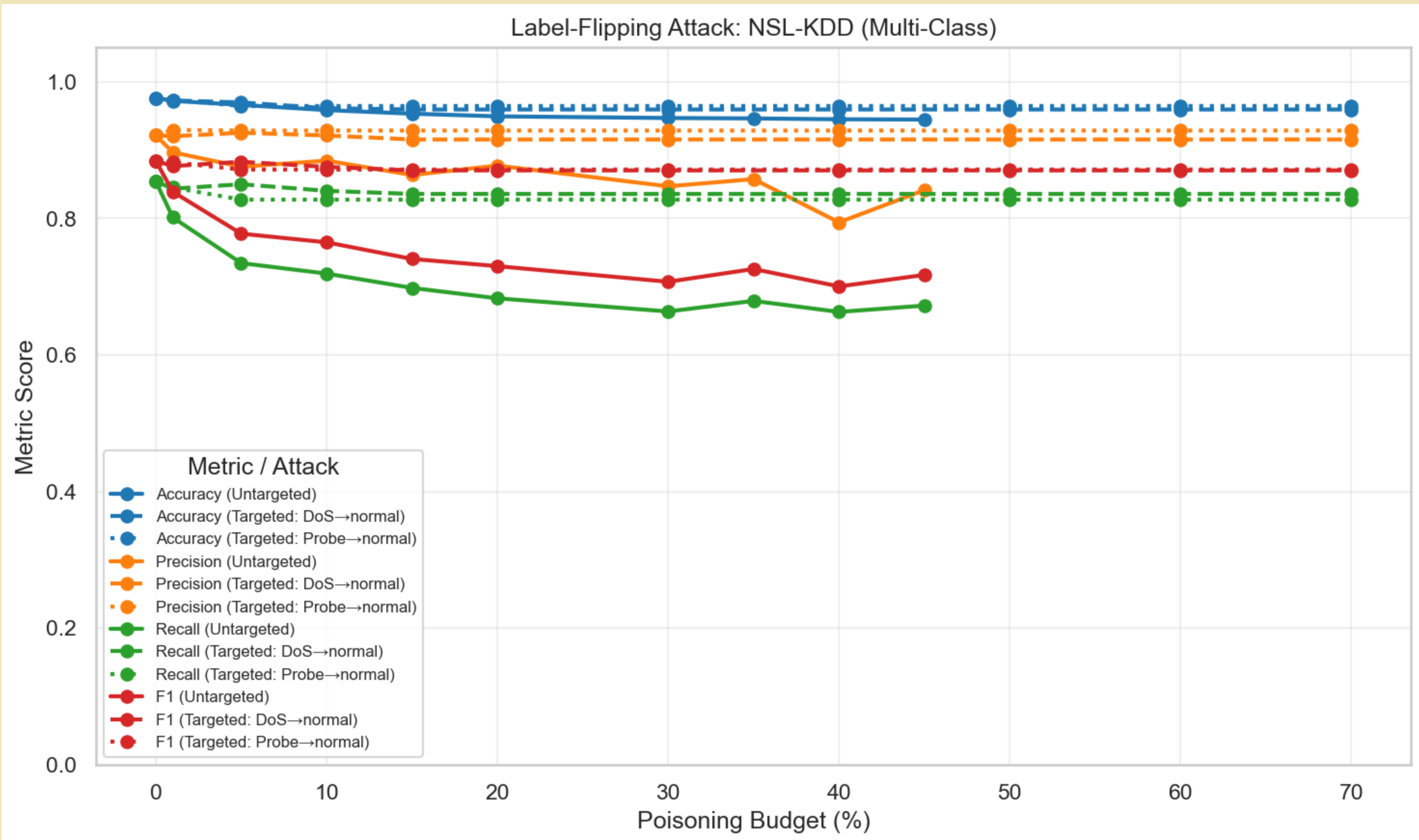
Results - Multiclass

Targeted (malware→benign) Budget: 0%					Targeted (malware→benign) Budget: 15%					Targeted (malware→benign) Budget: 30%				
Actual	benign	defacement	malware	phishing	Actual	benign	defacement	malware	phishing	Actual	benign	defacement	malware	phishing
	benign	defacement	malware	phishing		benign	defacement	malware	phishing		benign	defacement	malware	phishing
	Predicted					Predicted					Predicted			
benign	84208	48	39	1326	benign	84262	62	0	1297	benign	84262	62	0	1297
defacement	62	19075	7	148	defacement	110	19039	0	143	defacement	110	19039	0	143
malware	129	64	6037	274	malware	6012	86	0	406	malware	6012	86	0	406
phishing	2683	285	82	15772	phishing	3301	294	0	15227	phishing	3301	294	0	15227

Results



Results-Tabular Data



Results-Tabular Data

		Targeted (DoS→normal) Budget: 0%					Targeted (DoS→normal) Budget: 15%					Targeted (DoS→normal) Budget: 30%								
Actual	DoS	10606	9	0	0	62	DoS	10163	10	0	0	504	DoS	10163	10	0	0	504		
	Probe	25	2665	1	1	124	Probe	12	2627	1	0	176	Probe	12	2627	1	0	176		
	R2L	3	0	530	2	215	R2L	5	5	533	2	205	R2L	5	5	533	2	205		
	U2R	1	1	6	32	10	U2R	2	0	6	30	12	U2R	2	0	6	30	12		
	normal	91	78	82	4	15156	normal	55	133	80	4	15139	normal	55	133	80	4	15139		
		DoS	Probe	R2L	U2R	normal			DoS	Probe	R2L	U2R	normal			DoS	Probe	R2L	U2R	normal
		Predicted					Predicted					Predicted								

Conclusion



- (Spam-Ham) Precision and recall degrade asymmetrically (Random label flipping just causes the model to make more false-alarms, whereas targeted attacks lead to missed detection)
- Spam-Ham detection completely collapses after increasing the poisoning budget to 30%
- MURLs multi-class classifiers are immediately sensitive to poisoning (recall degradation from the very first budget step)
- MURLs Targeted attacks already reach the maximum damage already at just 5%, meaning maximum efficiency at low cost
- Targeted poisoning is budget-efficient but class dependent
- Attack impact depends more on source class characteristics than budget size



3

Reality

How realistic is this threat?



Poisoning in Reality



Grey/Black Box

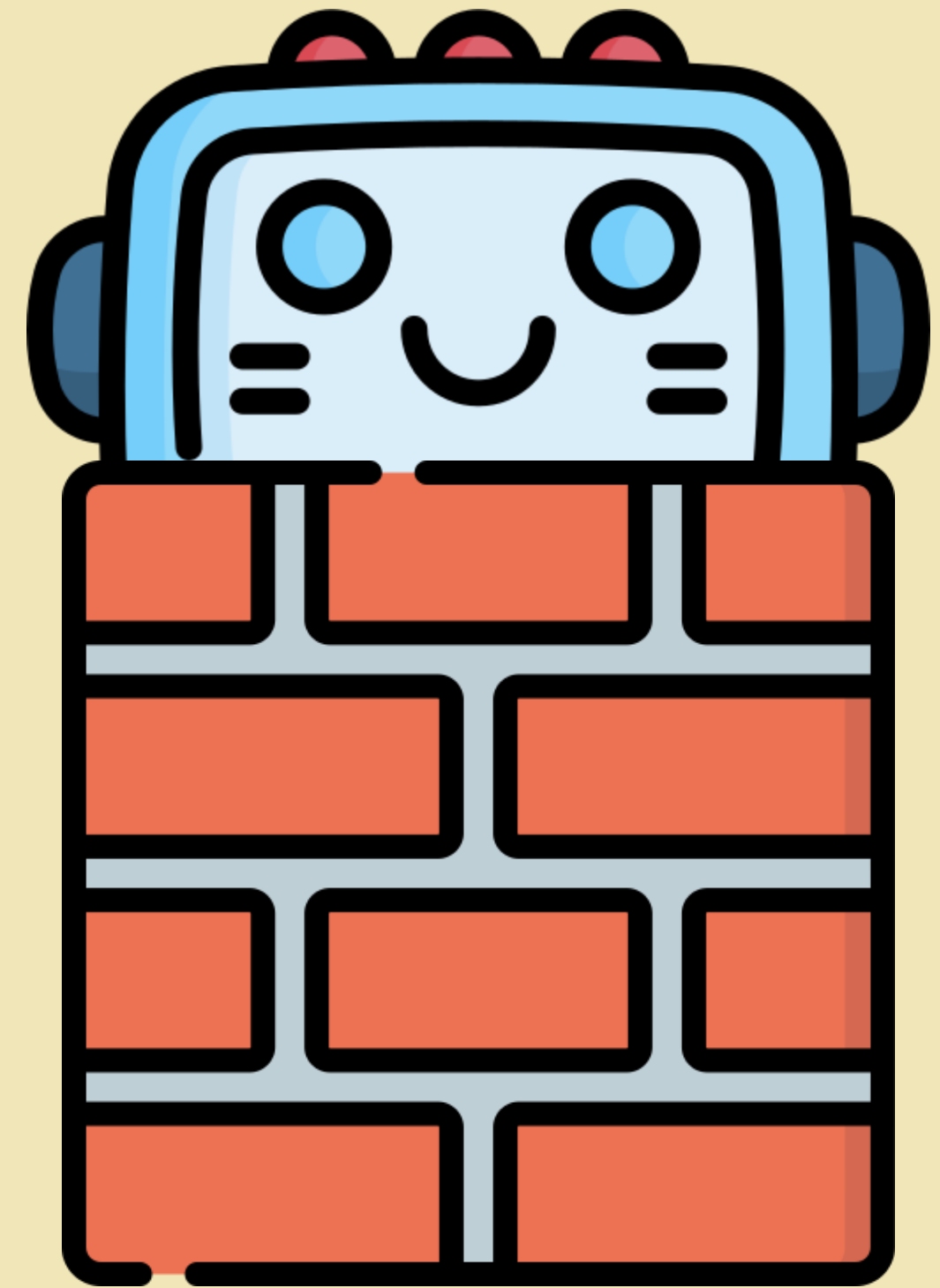
- Unknown Features Space, Algorithm – rely on proxy data
- Sanitization Defenses
- Merged datasets decrease impact

Detectability

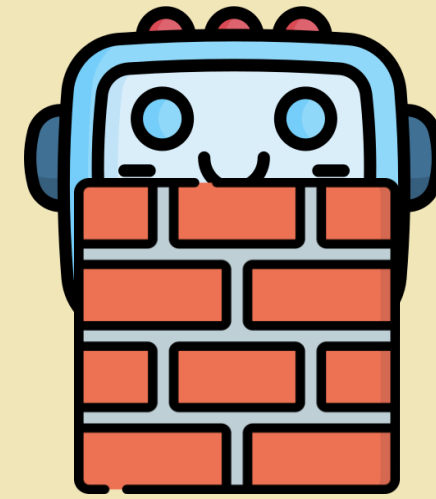
- Human-in-the-Loop
- Datasets are checked
- Decrease of metrics lead to suspicion

3

How to Defend



Defend



Training Data

Sanitization

RONI

Region of Non-Interests
exclude that decrease model
accuracy

(Nelson et al.)
Chan, P.P.K., He, ZM., Li, H. et al.

Label Sanitization

K-NN for
filtering or relabeling

Andrea Paudice, Luis Muñoz-
González, and Emil C. Lupu

Outlier Detection

Center distance
& designated spherical
radius

Steinhardt, Jacob, Pang Wei W. Koh,
and Percy S. Liang

Model Robustness

trimmed Loss-Function,
randomized smoothing,
adding noise





Thank you!

And keep your data safe!



References



Template: <https://www.slidescarnival.com/?q=AI>

Figures: <https://www.flaticon.com/>

Datasets:

- <https://www.kaggle.com/datasets/marcelwiechmann/enron-spam-data>
- <https://www.kaggle.com/datasets/sid321axn/malicious-urls-dataset>
- <https://www.kaggle.com/datasets/hassano6/nsllkdd>

Literature:

- Slides J. Wachter
- Chan, P.P.K., He, ZM., Li, H. *et al.* Data sanitization against adversarial label contamination based on data complexity. *Int. J. Mach. Learn. & Cyber.* **9**, 1039–1052 (2018). <https://doi.org/10.1007/s13042-016-0629-5>
- Paudice, Andrea, Luis Muñoz-González, and Emil C. Lupu. "Label sanitization against label flipping poisoning attacks." *Joint European conference on machine learning and knowledge discovery in databases*. Cham: Springer International Publishing, 2018.
- Steinhardt, Jacob, Pang Wei W. Koh, and Percy S. Liang. "Certified defenses for data poisoning attacks." *Advances in neural information processing systems* 30 (2017).